

Business Analyst Intern — Technical Assessment

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Question Set A — Answers

Section 1 — Data Quality Checks

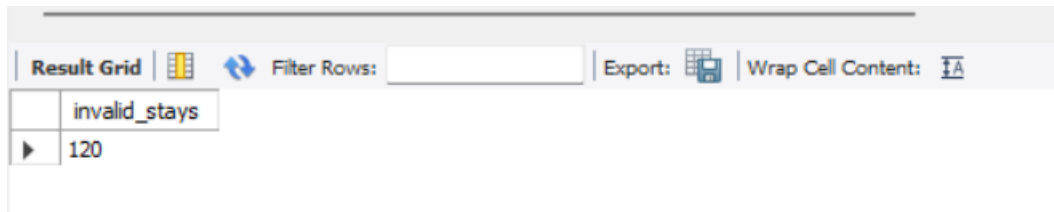
A1. Invalid stays (checkout_date ≤ check-in_date)

To identify data quality issues, booking records where the checkout date was earlier than or equal to the check-in date were classified as invalid stays.

SQL used:

```
select count(*) as invalid_stays from hotel_bookings where checkout_date <= checkin_date;
```

Answer: 120 bookings.



invalid_stays
120

These records represent data-quality issues and should be excluded from revenue and operational analyses because they do not represent valid hotel stays.

A2. Review Ratings by Customer Segment

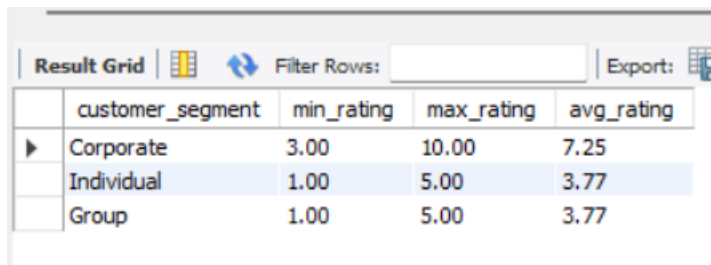
Review ratings were analyzed separately for each customer segment using the minimum, maximum, and average rating values.

The analysis showed the following results:

SQL used:

```
select customer_segment, min(cast(review_rating as decimal(5,2))) as min_rating,  
max(cast(review_rating as decimal(5,2))) as max_rating,  
round(avg(cast(review_rating as decimal(5,2))),2) as avg_rating  
from hotel_bookings where review_rating is not null and review_rating <> " group by  
customer_segment;
```

The analysis showed the following results:



	customer_segment	min_rating	max_rating	avg_rating
▶	Corporate	3.00	10.00	7.25
	Individual	1.00	5.00	3.77
	Group	1.00	5.00	3.77

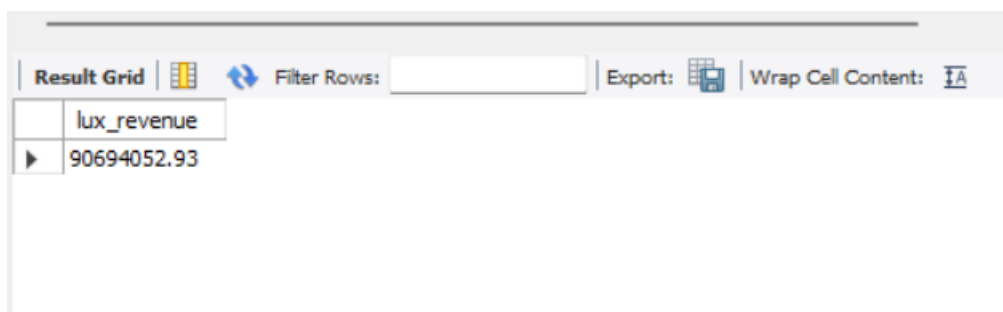
Implication: The segment with the highest average review rating demonstrates greater customer satisfaction and may exhibit stronger loyalty and repeat-booking behavior compared to other segments.

A3. Realized Revenue - Luxury Properties

After applying the required data-quality filters and considering only completed bookings with valid stay dates, realized revenue was calculated for Luxury properties.

SQL used:

```
select  
round(sum(total_amount),2) as lux_revenue  
from hotel_bookings where property_type = 'Luxury' and booking_status = 'Completed'  
and checkout_date > checkin_date and num_rooms > 0;
```



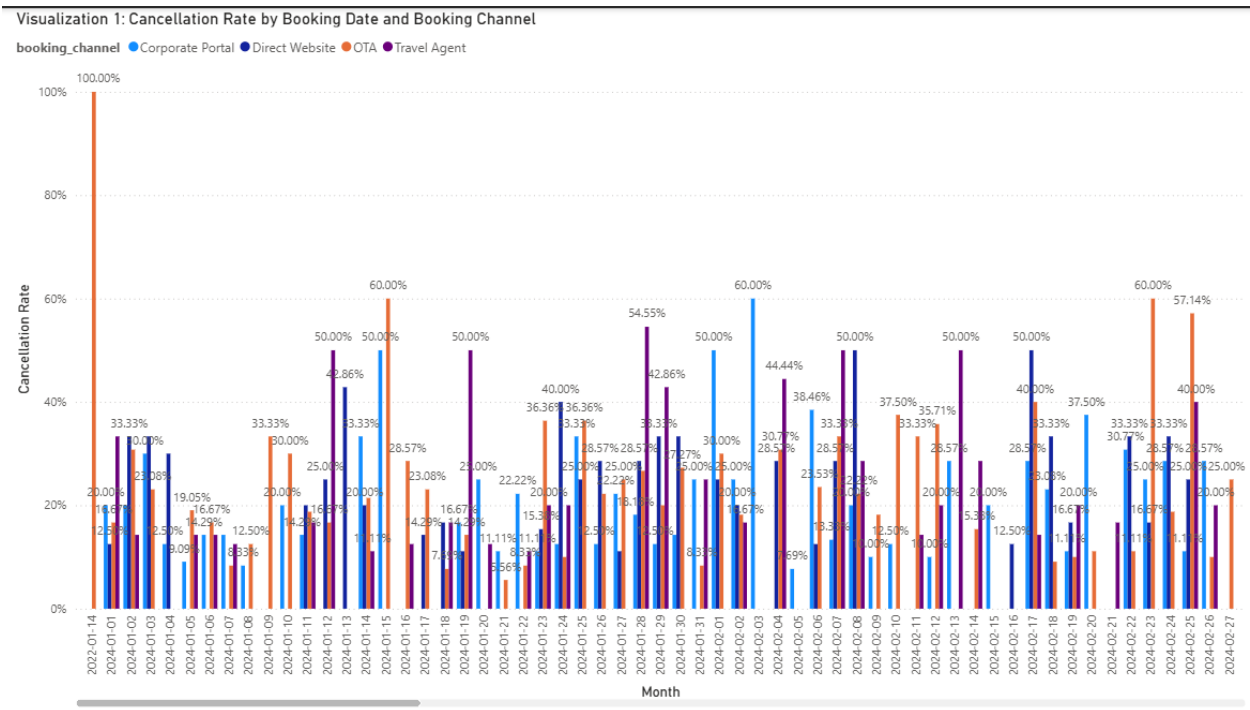
	lux_revenue
▶	90694052.93

This value represents actual recognized revenue after excluding cancelled bookings and invalid stay records.

Section 2 — Case Study: The Cancellation Crisis

A1. Cancellation Landscape

Cancellation Rate by Month and Booking Channel



The chart shows how cancellation rates differ across booking channels over time. It is clear that cancellation patterns vary significantly depending on the channel through which the booking was made suggesting that the booking source plays an important role in customer cancellation behavior.

Among all channels, Online Travel Agency (OTA) bookings consistently experienced the highest cancellation rates with some periods exceeding 40%. Bookings made through the Corporate Portal also showed relatively high cancellation levels. In contrast direct Website bookings generally had lower and more stable cancellation rates.

The differences in cancellation rates between channels are much more noticeable than the fluctuations within the same channel over time. This indicates that the booking channel is one of the key factors influencing whether a reservation is likely to be cancelled. Based on these findings OTA and Corporate Portal bookings appear to be the most cancellation-prone segments and warrant further analysis to identify the underlying causes.

A2. Rate vs. Volume Analysis

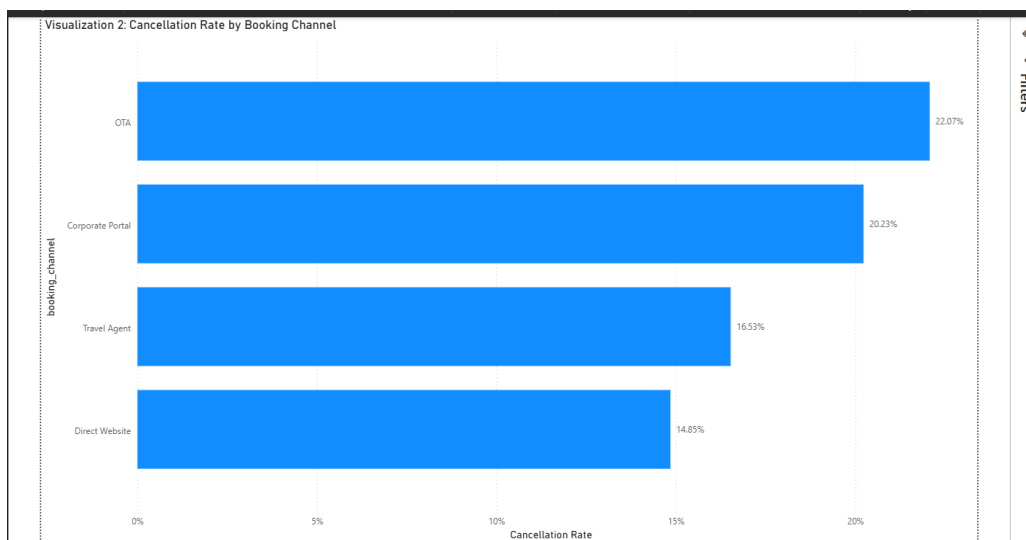
SQL Used:

```
select booking_channel, count(*) as total_bookings,  
sum(case when booking_status='Cancelled' then 1 else 0 end) as cancelled_bookings,  
round(sum(case when booking_status='Cancelled' then 1 else 0 end)*100.0/ count(*),2) as  
cancellation_rate  
from hotel_bookings group by booking_channel order by cancellation_rate desc;
```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content: IA
	booking_channel	total_bookings	cancelled_bookings	cancellation_rate
▶	OTA	4245	937	22.07
	Corporate Portal	3213	650	20.23
	Travel Agent	2414	399	16.53
	Direct Website	2128	316	14.85

The highest-rate slices are also the highest-volume slices. OTA bookings represent the largest source of cancellations, contributing 937 cancelled bookings at a cancellation rate of 22.07%. This distinction matters because reducing cancellations in high-volume segments produces a much larger impact on the platform-wide cancellation rate than focusing only on small segments with high cancellation percentages. Therefore, OTA and Corporate Portal channels should be prioritized to achieve the board's target of reducing cancellations by 5 percentage points.

Cancellation Rate by Booking Channel



The analysis reveals clear differences in cancellation rates across booking channels. OTA (Online Travel Agency) bookings have the highest cancellation rate at 22.07%, which is noticeably above the platform's overall average of around 19%. Corporate Portal bookings also show a relatively high cancellation rate of 20.23%.

On the other hand, Direct Website bookings perform the best, with a cancellation rate of 14.85%, closely aligning with the company's target of 14%. The gap of more than seven percentage points between OTA and Direct Website bookings highlights the impact that booking channels can have on customer cancellation behavior. Factors such as booking flexibility, customer intent, or channel-specific policies may be influencing these differences.

Given these findings, OTA bookings appear to be the biggest contributor to cancellations and should be the main focus of improvement efforts. Addressing the causes behind cancellations in this channel could play a significant role in helping the company achieve its goal of reducing overall cancellation rates by five percentage points.

A3. Root Cause: Hypothesis Testing

Worst-Performing Slice Identified

The OTA (Online Travel Agency) channel was identified as the worst-performing slice, with a cancellation rate of **22.07%**, significantly higher than the platform average of approximately 19%.

Result Grid				
		Filter Rows:	Export:	
				Wrap Cell Content:
	booking_channel	total_bookings	cancelled_bookings	cancellation_rate
▶	OTA	4245	937	22.07
	Corporate Portal	3213	650	20.23
	Travel Agent	2414	399	16.53
	Direct Website	2128	316	14.85

Hypothesis 1: Lead-Time Effect

To determine whether OTA cancellations are driven by customers booking further in advance, the average lead time (days between booking date and check-in date) was compared across channels.

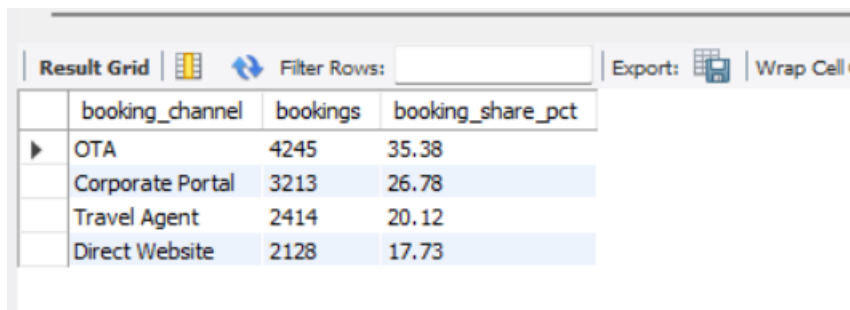
Result Grid		
		Filter Rows:
		Export:
		Wrap Cell Content:
	booking_channel	avg_lead_time_days
▶	Corporate Portal	33.27
	OTA	32.86
	Travel Agent	32.78
	Direct Website	30.55

Observation: OTA bookings have an average lead time of 32.86 days, which is nearly identical to Corporate Portal (33.27 days) and Travel Agent (32.78 days).

Conclusion: The lead time differences are minimal and do not explain OTA's significantly higher cancellation rate. Therefore, the lead-time hypothesis is not strongly supported by the data.

Hypothesis 2: Channel-Mix Effect

The booking share of each channel was analyzed to determine whether the platform's cancellation problem is concentrated within a specific acquisition channel.



The screenshot shows a software interface with a 'Result Grid' tab. Above the grid are controls for 'Filter Rows' and 'Export'. The grid itself has three columns: 'booking_channel', 'bookings', and 'booking_share_pct'. It contains four rows of data, with the first row expanded to show a sub-row.

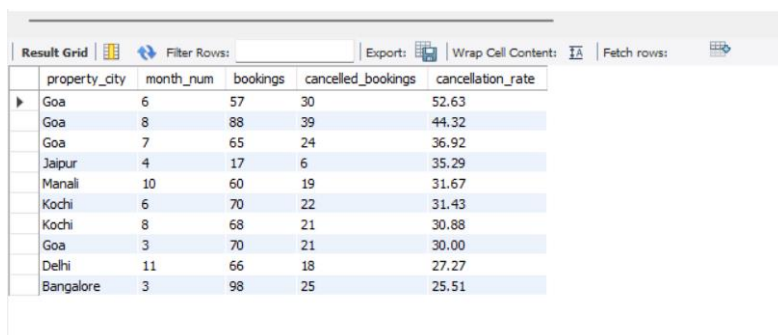
	booking_channel	bookings	booking_share_pct
▶	OTA	4245	35.38
	Corporate Portal	3213	26.78
	Travel Agent	2414	20.12
	Direct Website	2128	17.73

Observation: OTA contributes the largest share of platform bookings (35.38%) while also exhibiting the highest cancellation rate (22.07%).

Conclusion: The data strongly supports the channel-mix hypothesis. OTA bookings represent both the largest booking volume and the highest propensity to cancel, making this channel the primary contributor to platform-wide cancellations.

Hypothesis 3: City-and-Season Effect

Cancellation rates were analyzed by city and check-in month to determine whether seasonality contributes to cancellation behavior.



The screenshot shows a software interface with a 'Result Grid' tab. Above the grid are controls for 'Filter Rows', 'Export', 'Wrap Cell Contents', and 'Fetch rows'. The grid has five columns: 'property_city', 'month_num', 'bookings', 'cancelled_bookings', and 'cancellation_rate'. It contains ten rows of data.

	property_city	month_num	bookings	cancelled_bookings	cancellation_rate
▶	Goa	6	57	30	52.63
	Goa	8	88	39	44.32
	Goa	7	65	24	36.92
	Jaipur	4	17	6	35.29
	Manali	10	60	19	31.67
	Kochi	6	70	22	31.43
	Kochi	8	68	21	30.88
	Goa	3	70	21	30.00
	Delhi	11	66	18	27.27
	Bangalore	3	98	25	25.51

Observation: Goa experiences exceptionally high cancellation rates during the monsoon season (June–August), reaching as high as 52.63% in June.

Conclusion: City-season effects exist, particularly for Goa during monsoon months. However, these effects are limited to specific locations and periods, whereas OTA consistently demonstrates elevated cancellation rates across the entire platform.

A4. Targeted Recommendation

- **Who is targeted?**

Customers booking through the OTA channel with lead times greater than 30 days before check-in.

- **What is the policy?**

Introduce a partial non-refundable deposit of 10–15% for OTA bookings made more than 30 days before check-in. Customers who cancel would forfeit the deposit, while customers who complete their stay would have the deposit applied toward the final booking amount.

- **Why this segment?**

The OTA channel has the highest cancellation rate on the platform (22.07%) and accounts for the largest share of bookings (35.38%). Reducing cancellations within this segment provides the greatest opportunity to lower platform-wide cancellations.

- **Quantified Expected Impact**

The OTA channel is responsible for the highest cancellation rate on the platform (22.07%), with 937 cancellations from 4,245 bookings. To address this, a 10–15% non-refundable deposit can be introduced for OTA bookings made well in advance. Based on typical industry outcomes, this policy could reduce OTA cancellations by approximately 15–25%, preventing around 141–234 cancellations. Across the platform's 12,000 total bookings, this translates to a reduction of roughly 1.2–2.0 percentage points in the overall cancellation rate.

- **Expected Outcome**

The platform-wide cancellation rate could decrease from the current 19% to approximately 17.0–17.8%, making meaningful progress toward the board's target.

- **Risk**

Some customers may prefer flexible cancellation options and could be less likely to book under a stricter policy, resulting in a small decline in booking volume.

Section 3 — SQL Challenge

Schema Design

Three core tables (customers, properties, bookings) plus a reviews table.

- **Customer Table**

```
create table customers(  
  customer_id int primary key,  
  customer_name varchar(100) not null,  
  customer_segment varchar(20) not null,  
  customer_signup_date date not null,  
  customer_home_city varchar(50),  
  customer_loyalty_tier varchar(20));
```

- **Properties Table**

```
create table properties(  
  property_id int primary key,  
  property_name varchar(100) not null,  
  property_city varchar(50) not null,  
  property_type varchar(20),  
  property_star_rating decimal(2,1),  
  unique(property_name, property_city));
```

- **Booking Table**

```
create table bookings(  
  booking_id int primary key,  
  customer_id int not null,  
  property_id int not null,  
  booking_date date not null,  
  checkin_date date not null,  
  checkout_date date not null,  
  num_rooms int not null,  
  total_amount decimal(12,2) not null,  
  booking_channel varchar(30),  
  booking_status varchar(20),  
  foreign key(customer_id) references customers(customer_id),  
  foreign key(property_id) references properties(property_id),  
  check(checkout_date > checkin_date),  
  check(num_rooms > 0));
```


- **Review Table**

```
create table reviews(
review_id int auto_increment primary key,
booking_id int not null,
review_rating decimal(3,1),
foreign key(booking_id) references bookings(booking_id));
```

A1. Highest realized-revenue property per city

```
with property_revenue as(select property_city, property_id, property_name, sum(total_amount)
as revenue
from hotel_bookings where booking_status = 'Completed'
group by property_city, property_id, property_name), ranked_properties as (select *, rank()
over(partition by property_city
order by revenue desc) as rnk
from property_revenue) select property_city, property_id, property_name, revenue from
ranked_properties where rnk = 1;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	property_city	property_id	property_name	revenue
▶	Bangalore	2	The Grand Plaza	13957983.730000004
	Chennai	53	Saffron Heights	14355166.459999995
	Delhi	49	Mango Retreat	4200186.890000001
	Goa	23	Sapphire Retreat	19619575.909999996
	Jaipur	30	Ivory Retreat	1805426.609999999
	Kochi	1	The Grand Plaza	6631787.86
	Manali	10	Lotus Manor	4347636.540000001
	Mumbai	11	Coral Palace	10719830.160000004
	Pune	34	Velvet Palace	6062114.470000002
	Udaipur	6	Hilltop Inn	15669688.120000003

A2. Customers with average gap between Completed bookings < 30 days

```
with booking_gaps as(select customer_id, datediff(checkin_date, lag(checkin_date)
over(partition by customer_id order by checkin_date)) as gap_days
from hotel_bookings where booking_status = 'Completed'), avg_gap as (select customer_id,
avg(gap_days) as avg_gap_days
from booking_gaps where gap_days is not null group by customer_id)
select count(*) as customers_under_30_days from avg_gap where avg_gap_days < 30;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	customers_under_30_days			
▶	445			

Section 4 — Mini-Project (Weather-Cancellation Analyzer)

- **Objective**

The objective of this mini-project was to enrich the hotel booking dataset with external weather data and determine whether adverse weather conditions are associated with higher cancellation or no-show rates.

API Used:

Open-Meteo Historical Weather API (Free, No API Key Required)

Endpoint:

<https://archive-api.open-meteo.com/v1/era5>

Cities analyzed:

- Goa
- Manali

All Goa and Manali bookings with check-in dates in 2024.

API Integration & Error Handling

The notebook makes live API calls to Open-Meteo and includes exception handling and request timeouts to ensure the program does not fail when the API is unavailable or returns incomplete data. Weather data is fetched once per city for the full date range rather than once per booking, minimizing API calls and improving efficiency.

Data Merge

Weather data was merged back to the booking dataset using:

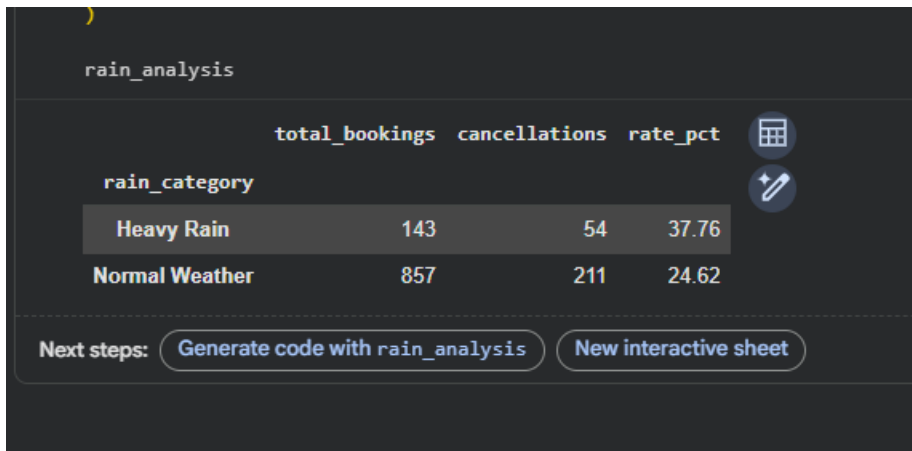
- `Property_city` -- city
- `Checkin_date` -- weather date

This ensured each booking was assigned the actual weather conditions observed on its check-in date.

Quantified Insight

Bookings were classified into two categories:

- Heavy Rain: Daily precipitation ≥ 20 mm
- Normal Weather: Daily precipitation < 20 mm



	total_bookings	cancellations	rate_pct
rain_category			
Heavy Rain	143	54	37.76
Normal Weather	857	211	24.62

Next steps: [Generate code with rain_analysis](#) [New interactive sheet](#)

Heavy-rain days recorded a cancellation/no-show rate of 37.76%, compared with 24.62% during normal weather conditions. This is a difference of 13.14 percentage points, indicating that bookings were approximately 53% more likely to be cancelled or result in a no-show during periods of heavy rainfall. This insight could not have been derived from the booking dataset alone and required external weather information.

README

- This notebook combines hotel booking data with historical weather data from the Open-Meteo API.
- It retrieves daily precipitation and temperature information for Goa and Manali during 2024.
- Weather records are merged with bookings using city and check-in date.
- Bookings are categorized into Heavy Rain and Normal Weather groups.

- Cancellation and no-show rates are compared across weather conditions.
- Run all notebook cells after uploading the exported CSV file.

AI Usage Note

AI was used to generate python codes like API integration, data retrieval, weather categorization.

The generated code required modification to match the actual dataset structure.

Merge keys were corrected to use property_city and checkin_date.

Date formats were standardized before joining datasets.

Error handling and timeout logic were verified and adjusted to ensure successful execution.

Final calculations and interpretations were validated using the generated results.

Simply we say AI used for python codes.

END